# Predicting the CPI to Median Income Ratio in America

Jarvis Sittel

**Table of Contents**

Introduction: 1

Methodology: 3

Results and Discussions: 5

Conclusion: 7

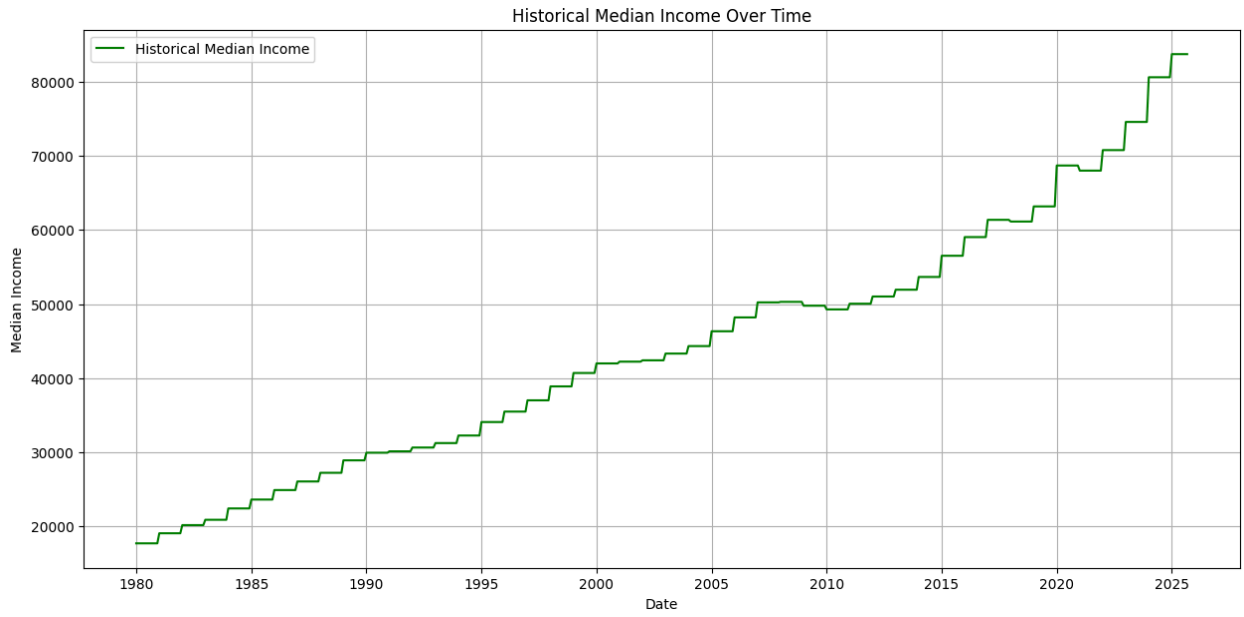
Citations: 8

**Abstract**

The CPI to Median income ratio shows a slow degradation of American wages. Using an LSTM encoder-decoder model, it was shown that in the next twenty years we would lose approximately another thirteen percent. This is incredibly important for governing bodies and economic players to understand as we enter a future with less disposable spending money from middle and lower income families. The data shown in this report is only a small piece of a much bigger pie and should be combined with other similar metrics to form a bigger picture.

## Introduction

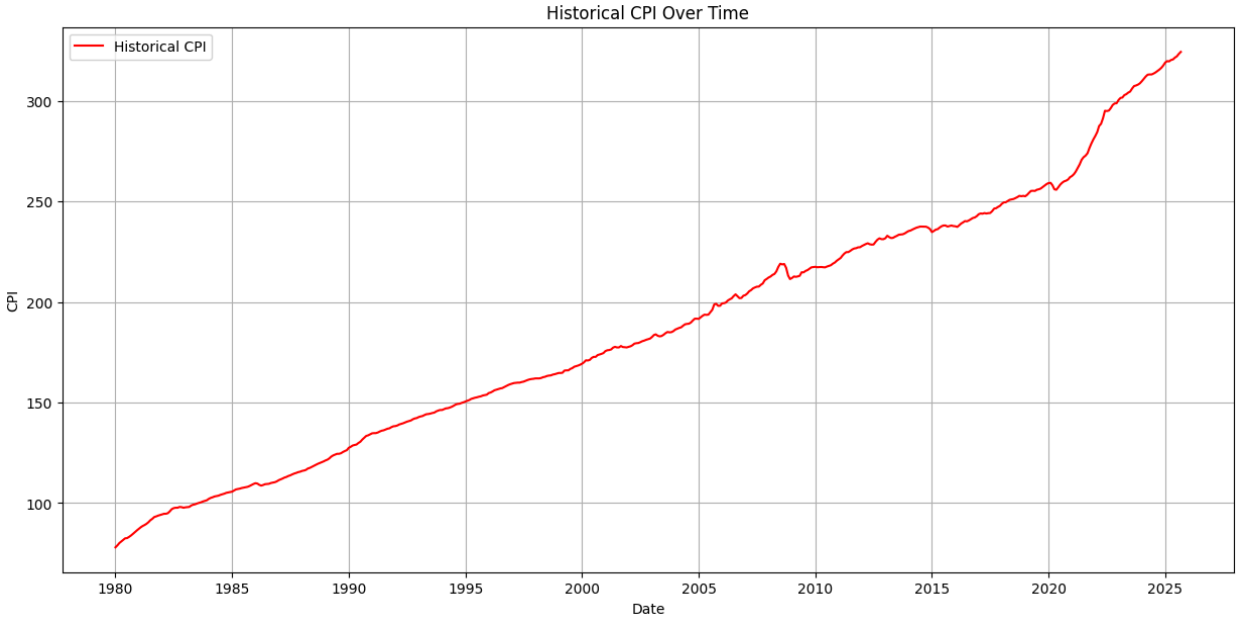
The median household income is not the end all be all of indicators for people's economic situations in the United States, but it is one of the best ones we have to look at. Utilizing median household income and a consumer price index, this project hopes to look at the rate that inflation is out pacing people's wages. Inflation outpacing the median income results in a significant decline in the quality of living for Americans. Inflation outpacing rising wages means that people will struggle to maintain their current lifestyles, and could be very bad for the economy as a whole. Many industries would  The project intends to predict the rate at which this erosion of the median income by inflation will happen.




Utilizing RNNs to predict the degradation of the buying power of the median American allows us to utilize the advancements in deep learning technology to make informed decisions about the

economy. Preparing for a world where the median American can no longer afford to live due to price increases is an unfortunate but necessary exercise.

There was only a single study that was very similar to this one and a few that were related. What I did find was that Gutierrez-Fierros found that CPI did have an impact on Americans and their ability to save, but that it was not major from 2013 to 2023, but according to my ratio, the difference from 1980 to 2025 has been far more drastic. What this really means is that over the course of the next few decades, we could be looking at a very real reduction in the spending power of median American salaries due to being unable to keep up with inflation.




Shorrocks and Marlin talk in detail about how inflation affects income, but they aren't as focused on predicting future deterioration. This report will be focused on predicting the future, so law makers can write legislation or adjust rates and individual citizens can plan, save, and vote accordingly.

**Methodology**

The data used as the target variable for the model is a ratio of CPI from FRED and median household income from the United States' Census Bureau. The model was trained on that variable, the median household income, and the CPI. Poverty rate, a wealth inequality metric, and a few other variables were considered to add complexity to the model, but they each had downsides that seemed to negatively affect the model. It's discussed further later into this section, but the model struggled to predict cycles. To remedy this, four extra data sources were used to help the model see the cyclical nature of the economy, but it never caught on.

| Layer (type) | Output Shape | Param # |
|---|---|---|
| input_layer (InputLayer) | (None, 20, 3) | 0 |
| lstm (LSTM) | (None, 100) | 41,600 |
| repeat_vector (RepeatVector) | (None, 1, 100) | 0 |
| lstm_1 (LSTM) | (None, 1, 100) | 80,400 |
| time_distributed (TimeDistributed) | (None, 1, 1) | 101 |

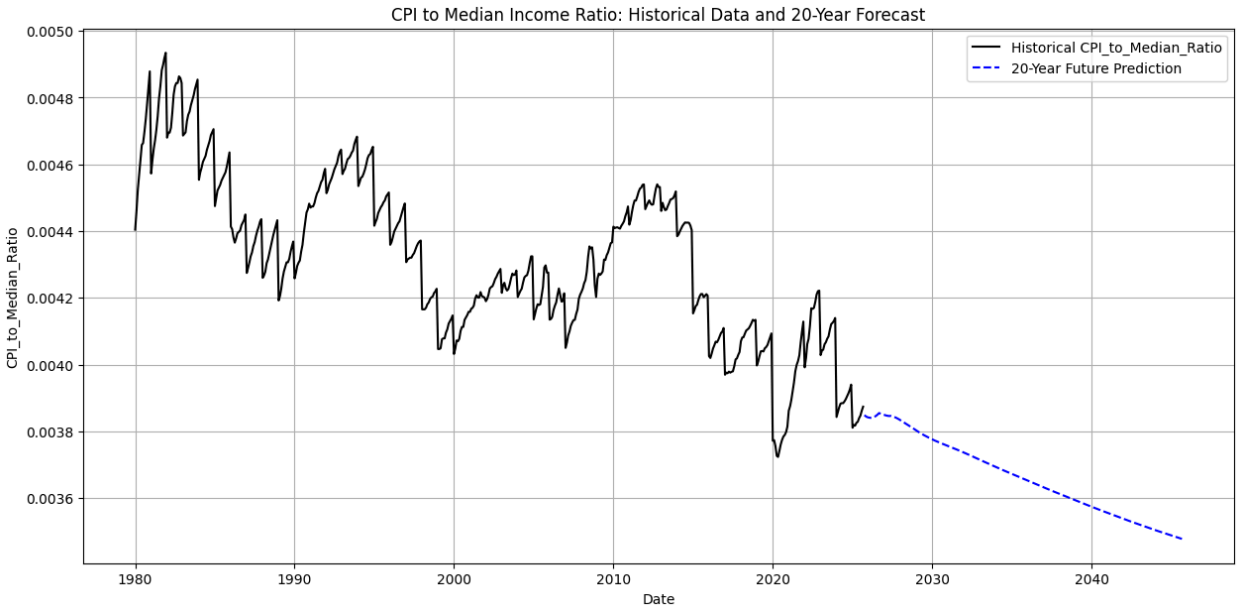
An LSTM encoder-decoder model was chosen as it was designed to take time series and use recursive forecasting to create a sequence of future values. This is exactly what was needed to predict the future degradation of the median income. Also, having tried many different data sources and LSTM made the most sense as they are rather flexible. The disadvantages lie in the way that it predicted the values. The LSTM encoder-decoder method refused to predict values in a cyclical way. The previous data shows a clear cycle to the ratio as we enter recessions and periods of growth, the value ebbs and flows, but the model refused to show that as the case. I am unsure if other models would have been able to, but LSTMs are known for their lack of

interpretability. This was very apparent in training the model. LSTMs are also known to need large amounts of data which was not present. Based on its data requirements, expectations for the model were rather low, but it managed to predict in a completely reasonable way. The model is predicting a single value, so it really did not need to be very complex. The predictions use iterative forecasting which was implemented with the help of Google's Gemini. After hours of running the same code in circles without any successful attempts, the assistance was greatly needed.

In the end, the model was trained on the historical data of CPI, median income, and the CPI to median income ratio. This allowed it to get an idea of the relationship between the three then the predictions are based on those previous findings and projected values for CPI and median income. This is what resulted in the straight line rather than the cyclical values that would have been expected. There were ways to remedy this, but they went a little over my head. I opted against using things I could not understand simply because Gemini thought it would be best. The values are just a percent change based on the previous months. The predicted values are scaled and fed to the model which then uses an iterative forecasting loop to predict future values. The model did very well and managed a loss of less than a percent through most of its training. The future values for CPI and median income are the weakest link, but given historical trends, the model wound up predicting exactly what was expected.

**Results and Discussions**

The results were exactly as expected. The model struggled to predict in a cyclical pattern, but it showed a steady decline matching the previous data. According to the model, the median income will lose approximately twelve percent of its value to inflation over the next twenty years. The median income and CPI would rise at differing rates which results in a loss of relative income for many Americans. This is only a single example of what could happen as the economy is a very large and ever changing beast.




The effects of Americans losing twelve percent of their income would be devastating to the economy which is what the numbers say has happened in the past, but CPI is not an end all metric. A governing body who has more information would be much more equipped to make long term judgment calls. Americans losing twelve percent of their buying power would hurt the economy at large by slowing spending, slowing GDP growth, and stifling industries that require excess spending money such as tourism. More importantly, the effects would be felt by

individuals living in lower income brackets. People in the median may still be able to get by, but those on the line between livability and poverty may find themselves struggling to stay afloat.

The government or adjacent bodies would be remiss to ignore data that so blatantly shows a scary fate for many Americans which means they are likely already aware. Even if my model may not be the most reliable source as it took a long time and Google Gemini to get it to output anything that made sense, the trend is very real. The Fed or some governing body should keep their eyes on the future of the median income's ability to be livable. As this ratio decreases many Americans will start spending less, but eventually the scale at which Americans struggle to afford groceries, toiletries, and other necessities will grow thanks to ballooning prices. This model and many like it could be a great aide to governing bodies when looking toward the future. Understanding the importance of middle and lower class Americans' spending power is a necessity for those who have influence over the economy.

**Conclusions**

This paper stands as an example of the kind of research that could be important for the Federal Reserve, or some other governing body. Looking at data like this, rising poverty rates, and other similar data, the U.S. government could decide on interest rates to help stifle inflation, allot funding to programs that help people who are falling behind thanks to their wages being cut, or adjust tax rates for middle and lower income families that have lost a significant amount of spending power. This project specifically shows one possible outcome given historical data. There is no guarantee that things will continue exactly as they have, but the model and precedent say they will. The economy, incomes, inflation, and a million other factors come into play. In the future, further research should be the result of many models studying a multitude of factors, so a bigger picture can be formed from the results.

The model is not perfect, there are multiple places where it could be improved from the complexity to the way that the future values for median and CPI are handled. That is to say that it may not be the end all be all, but it does highlight an important issue that the Fed or other governing bodies probably are and should continue to monitor going forward. Inflation is only one of many economic factors, so this model should not be the only thing that influences decisions. It could be part of a much larger collective of models or data sources that help inform decision makers.

**Citations**

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Prompt: “I have been struggling to get my predictions to make sense given historical trends, would you help suggest a prediction method?”